

OPC Code Atlas

How the repository builds a simulated recommendation world in which the logger sees users and items through distorted vectors, learns new policies from the logger's data, and grades them against the hidden truth. The two sections this revision centers on are the kinds of representation bias the study covers and the mechanisms that can correct each one. Since 4 Oct the logs are corrected: a logging bug (each user's action reused the draw that picked the user) was fixed, and every result was rerun and re-tuned on the corrected logs (`docs/simulator_fix_opc_revalidation_20261004.md`). On 5 Oct the CausE baseline was compared with the revalidated OPC at equal budget and equal capacity (`docs/cause_fair_comparison_25k.md`), and BLOB was reproduced and compared given the same supplied source (`docs/blob_controlled_integration.md`); then BLOB's prior was calibrated for catalog size (`docs/blob_prior_calibration.md`). On 6 Oct the compared arms' training objectives were audited, the live workflows were reviewed, and their bugs were fixed: rerunning, resuming and extending a run are now idempotent (`docs/training_objectives_audit.md`, `docs/handoff_20261006.md`). On 7 Oct three training objectives (penalized likelihood, propensity-weighted likelihood, OPC's value objective) were compared on one shared correction model, with the same regularizer, data and search (`docs/shared_objective_study.md`).

Repo `TauRoeed/OPC`, branches `CRM`, `cause-fair`, `blob-controlled-integration`, `blob-prior-calibration`, `handoff-20261006` (the production / handoff checkpoint) and `representation-mismatch-research-next` (new research)

Code at `24a4afa` (pushed, branch `representation-mismatch-research-next`; the handoff checkpoint `handoff-20261006` is unchanged at `9445a12`), 7 Oct 2026

Functions are named, not line-numbered Paths are relative to the repo root

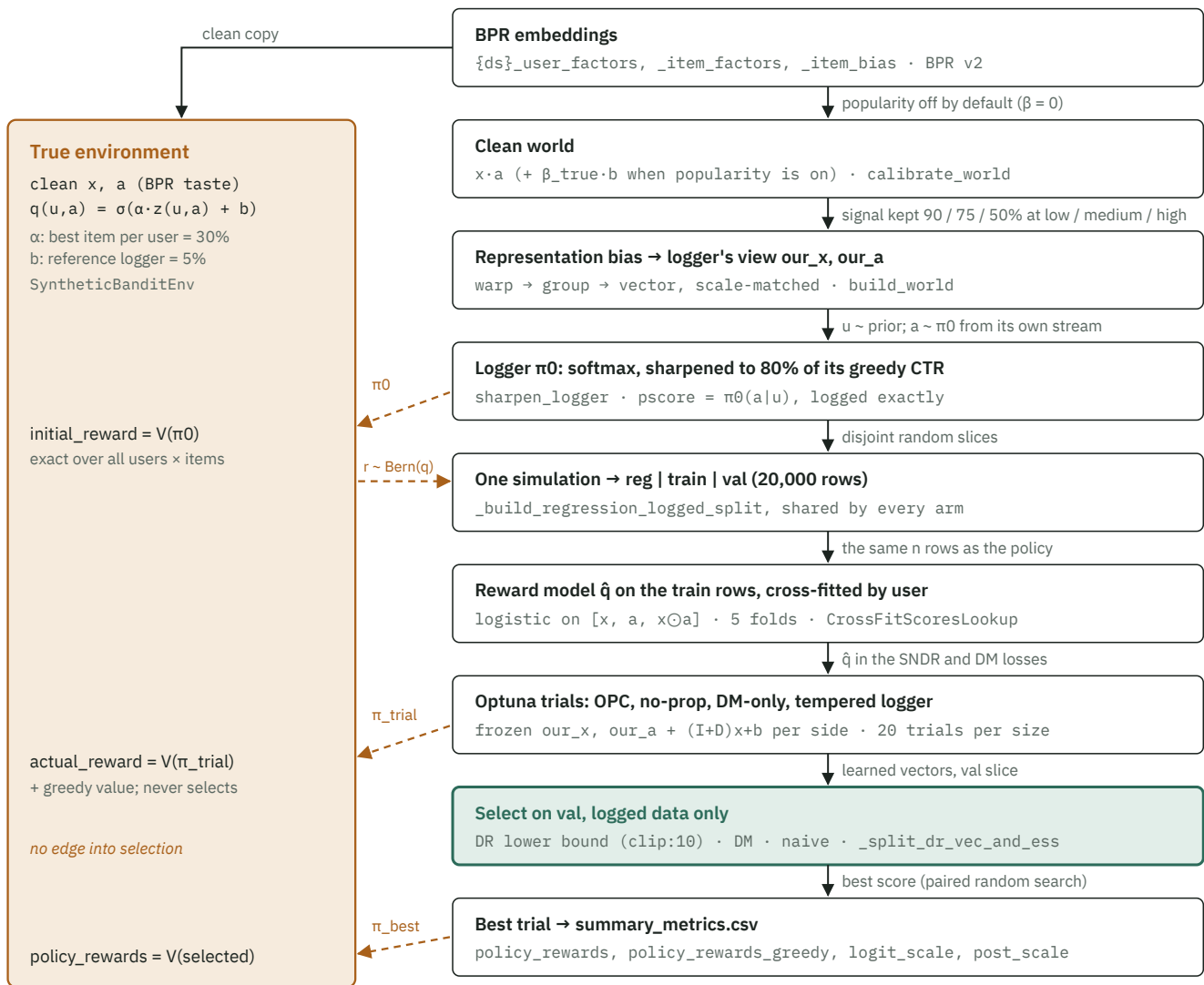
THE RESEARCH QUESTION

Can off-policy learning undo a logger's representation bias?

A deployed recommender scores items with learned user and item vectors. Those vectors are wrong in structured ways: a global misalignment of taste axes, whole user segments misread, individual tastes never learned. The logs it collects inherit those errors. The study asks which of these distortions a policy learned from the logs can correct, with which mechanism, how much data it takes, and whether logged propensities add anything beyond a good reward model. The simulator makes every distortion a dial and computes every policy's true click rate exactly.

The pipeline at a glance

A study is a grid of *conditions* (dataset × bias configuration × seed, run at several training sizes). Each condition runs the chain below for every arm. The design rests on the wall between the two columns: the true environment produces the rewards and grades the results, and nothing that chooses a policy ever reads it.



Solid arrows are the learning path; dashed amber arrows touch the hidden truth. The truth supplies the logged rewards and grades three things (the logger, every trial, the selected policy), but nothing flows from it into selection (green), which reads only the validation rows of the log.

FOCUS 1

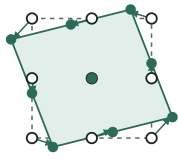
Representation bias: the distortions we cover

The truth is BPR's clean user and item vectors. The logger, the reward model and every learned policy see biased copies instead. Three kinds of bias are mixed in, in this order, on both sides (users and items), each rescaled to the clean vectors' size so a level means a share of signal replaced and never an inflation of vector length:

$\text{biased} = (1 - \epsilon) \cdot \text{current} + \epsilon \cdot \text{target}$, applied for warp, then group, then vector, and rescaled to the clean RMS after each step (`utils/representation_bias.py · SideBias.biased`).

Warp

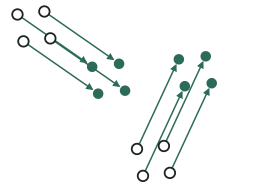
one linear map for the whole side



Taste axes misaligned for everyone:
a stale or globally mis-trained model
target = $x \cdot W$, $W \sim N(0, 1)^{(d \times d)}$

Group

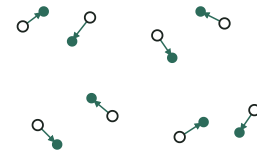
one offset per group of users or items



segment A segment B
Whole segments' tastes misread
target = $\text{template}[\text{group}(x)]$, k-means

Vector

independent noise per user and item



Individual tastes never learned:
sparse users, cold items
target = $N(0, I)$ per vector

Open circles are clean vectors, filled green dots the logger's biased view. A warp moves everyone by one shared map, so its structure is global. A group offset moves a whole segment together. Vector noise moves each user or item on its own, so no shared structure is left to learn from.

Levels, configurations and calibration

The strength ϵ of each type is calibrated per dataset so that **all three together** keep 90%, 75% or 50% of the taste signal at low, medium and high bias. Signal kept is the mean, over users, of the correlation across items between biased and clean taste scores. A single type at level L alone keeps κ_L of the signal (about 0.97, 0.94 and 0.88). A configuration names the three levels, for example `high/none/none` for warp only.

CONFIGURATION	WHAT IT ISOLATES	LOGGER'S GREEDY RANKING REACHES (SHARE OF THE 30% CEILING, ML / KUAIRAND / ANIME)	COVERED IN
<code>none</code>	control: no representation bias, only the logger's exploration	100%	dev grid, final report
<code>low · medium · high</code>	all three types together, 90 / 75 / 50% signal kept	80–89% · 59–73% · 38–47%	Test 2 (medium, high); final report (all)
<code>high/none/none</code>	warp only	73% / 82% / 73%	dev grid, final report
<code>none/high/none</code>	group only	77% / 87% / 79%	dev grid, final report
<code>none/none/high</code>	vector only	72% / 80% / —	dev grid, final report
<code>--pop-strength</code> , <code>--logger-pop-strength</code>	popularity: the truth weighs BPR's item bias b by β_{true} , the logger by β_{log} (over-exposure when $\beta_{\text{log}} > \beta_{\text{true}}$)	—	PLANNED RUNS (the dial exists)

The logger that collects the data

A softmax over the biased scores, sharpened per condition until it earns **80% of its own greedy CTR** (`--logger-greedy-share 0.8`, `sharpen_logger`). At that setting its temperature is about $T/4.2$ and it spreads over 110–260 items per user. It mostly exploits its ranking, which the bias has made wrong, and explores near the top of it. The spread temperature T , at which the clean logger covers half the catalog, only calibrates the click model. `off` restores the older near-uniform logger, which earned about 26% of its greedy value.

FOCUS 2

Correction mechanisms

A learned policy can fight the bias at three points: in what the policy can represent, in the objective it is trained on, and in how one of many trained candidates is chosen. Chips mark what exists today and what is planned.

1 · The policy class

What distortion of the vectors can the learned policy undo?

Linear correction

DEFAULT

(I + D)·x + b on the user side and on the item side, starting exactly at the logger (D = 0, b = 0). It can invert a warp exactly and remove a shared shift.

```
--policy-transform linear ·  
GlobalLinearCorrection
```

Linear + MLP

OPTION

The linear correction plus a zero-initialized MLP residual on the raw vector, with no LayerNorm and no dropout. Can bend the space nonlinearly, for example around group boundaries.

```
--policy-transform linear+mlp
```

MLP (older)

LEGACY

x + MLP(LayerNorm(x)), random start. Kept to reproduce earlier runs.

```
--policy-transform mlp
```

Sharpness: logit scale, post-hoc tempering

OPTIONS

A learnable s in softmax(s·u·a/T), or s chosen after training on validation from 0.25–16. These change how concentrated the policy is, never its ranking. The study learns s (re-checked on corrected logs: a fixed scale loses 0.2–1.3 points, and tempering on top does not change the selected policy).

```
--learn-logit-scale · --post-temper
```

Learnable popularity weight

AUTO

With a popularity column, the policy learns its own weight on b, starting at the logger's. This corrects over- or under-exposure.

```
CFModel(pop_weight)
```

Per-group and per-item offsets

PLANNED

A learned offset per user group (k-means on the biased vectors) and per item, pulled toward zero with strength shrinking as the group's or item's logged data grow (partial pooling). Aimed at group bias and at the part of vector bias that has enough data behind it.

2 · The training signal

How is a biased log turned into an objective?

Propensity correction (OPC)

DEFAULT ARM

The reward model's value of the policy plus an importance-weighted correction on the logged rows. This removes the logger's exposure bias where the logger explored. The working development default is DR, differentiated directly, with harmonic:0.1 weights; selection stays clip:10 with the 95% lower bound. Set on 27 Sep (f5cade9) and re-tuned on the corrected logs on 4 Oct. It is the working method, not the final paper choice. Exact SNDR (the gradient of the full-data ratio) is the legitimate self-normalized alternative; on corrected logs it adds nothing over DR with harmonic weights. Legacy minibatch SNDR, global SNDR (a stop-gradient surrogate) and the log trick remain for reproducibility only.

```
--policy-losses dr (default) | sn dr  
· --sn-scope exact|batch|global · -  
-opc-gradient direct (default) |  
log-trick
```

Importance-weight transform

OPTIONS

The options are none (raw DR), clip:M, shrink:λ (Su et al. 2020) and harmonic:λ (Metelli et al. 2021): w/(1 - λ + λw), at most 1/λ, and trained by its direct gradient. Under the log trick, a transform g optimizes H(w) = ∫ g(t)/t dt rather than the named estimate; for shrink:100 that is 10·arctan(w/10). On corrected logs every regularized transform beats raw DR on average (the best by about 0.3–0.5 points per cell), and the best amount of regularization falls with n. harmonic 0.1–0.3 lead; harmonic:0.1 is the most robust across sizes and stays the default. shrink:100 is within noise of it and is the robustness alternative; none (raw DR) is the reference.

```
--train-weights (default  
harmonic:0.1) ·  
docs/training_losses.md §3.4, §9
```

Reward model alone (DM-only)

BASELINE

Maximizes the policy's value under $\hat{q}$. It reaches items the logger never showed if $\hat{q}$ extrapolates well; no propensities.

```
dm loss · --methods dm
```

Logged clicks as they are (no-propensity)

BASELINE

Raises the probability of clicked logged items. It inherits the logger's exposure (it favors $\pi_0 \times q$) and stays near the logger.

3 · Selection

Which of 20 trained candidates is deployed?

DR lower confidence bound

OPC

Mean - t·SE of the doubly robust estimate on 20,000 validation rows, weights clipped at 10. On corrected logs it picks within 0.03–0.11 points of the best of 20 trials; raw-weight selection loses up to 0.8 points.

```
--select-weights clip:10 · --  
optuna-selection ci_low
```

Reward-model score

DM-ONLY

The DR score with every weight 0. select_estimator="dm"

Naive score

NO-PROPENSITY

Mean of r·π on validation.

Tempered logger

BASELINE

No training at all: the logger's logits × s, with s chosen by the DR score. It measures how much of any gain is sharpening alone.

```
--methods tempered_logger
```

Every rule, after the fact

LOGGED

Every trial of every arm carries a selection score per logged rule (19 in the revalidation runs: 18 weight transforms and DM), so any selection rule can be replayed on any training rule without new runs.

```
--log-select-weights
```

naive loss

The reward model $\hat{q}$ DEFAULT

Logistic regression on $[x, a, x \odot a]$ of the biased vectors, so rankings can differ between users. It is fit on the same n training rows as the policy and cross-fitted by user (5 folds).

```
--reward-features interaction · --
reward-data train · --crossfit-
folds 5
```

Bilinear reward model $x^T W a$ PLANNED

A full (ridge or low-rank) bilinear form represents a warp exactly; $x \odot a$ captures only its diagonal.

Cluster-level weights PLANNED

Importance weights over item clusters (MIPS, OffCEM) or a two-stage policy (POTEC), for catalogs where item-level weights collapse (msd, lastfm).

FOCUS 3

Which mechanism can fix which bias

Each cell rates a correction mechanism against a bias type, by construction and by the oracle bound. The oracle trains the policy class directly on the true click probabilities, with unlimited data (ml, high level of each type alone).

BIAS TYPE	LINEAR CORRECTION	GROUP / ITEM OFFSETS	REWARD MODEL $x \odot a$	BILINEAR REWARD MODEL	PROPENSITY CORRECTION	ORACLE BOUND, ML (GREEDY LOGGER → LINEAR → + GROUP OFFSETS)
Warp	exact	not needed	diagonal only	exact · planned	where π_0 explored	22.1 → 29.7 → 29.2 (ceiling 30.4)
Group	shared shift only	targets it · planned	partial	partial	where π_0 explored	23.5 → 28.2 → 28.9
Vector	de-noising only	items with data · planned	no structure	no structure	where π_0 explored	21.8 → 24.9 → 26.3
Popularity exposure	—	—	sees b	sees b	removes exposure	planned runs; policy learns its own weight on b
Logger sharpness (not a bias)	logit scale or post-hoc tempering; the tempered-logger baseline measures this part				noisy for flat loggers	sharpening alone: logger 15.5 → 19.3 (medium)

Warp is almost fully fixable by the linear class (7.6 of 8.3 lost points back), group bias mostly, vector noise only partly. Propensities help wherever the logger's exploration reached, so a sharper logger helps less at small data (see Findings). The dev grid measures these cells with actual learning: linear vs linear+MLP vs MLP, with and without post-hoc tempering, for each type alone.

The four arms

All arms get the same splits, the same reward model, the same search budget (20 trials per size), the same seed and, where they apply, the same weight rules. What differs is how each turns the log into an objective and a selection score.

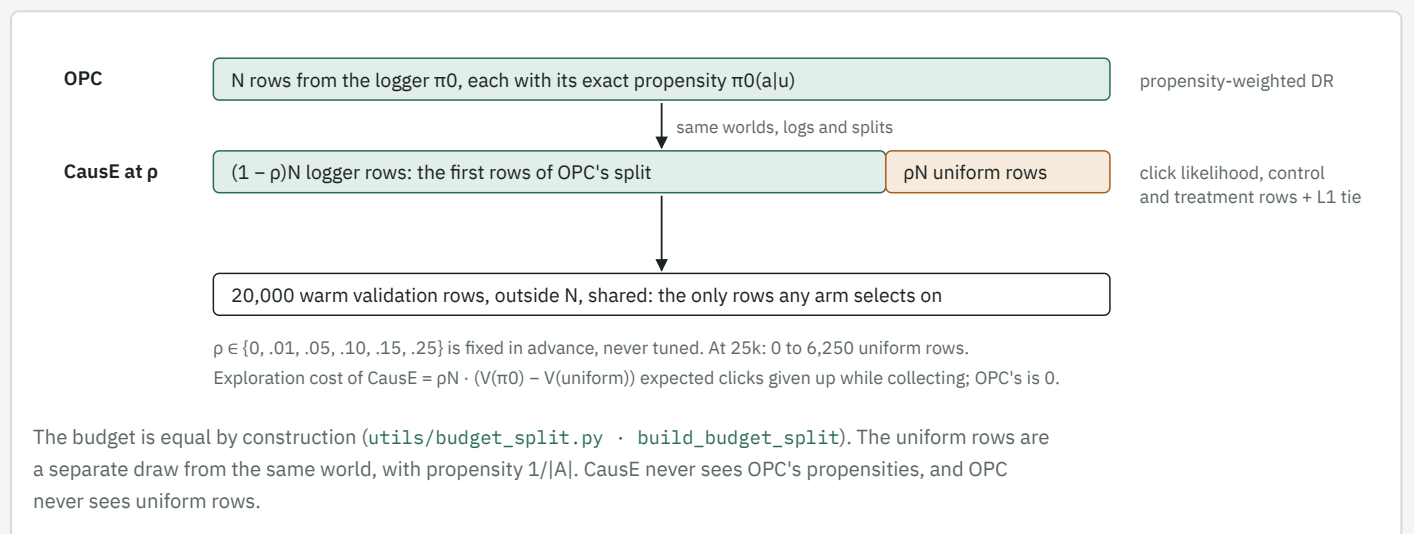
ARM	TRAINING OBJECTIVE	SELECTION SCORE	WHAT IT ISOLATES
OPC DEFAULT	DM($\hat{q}$) + importance-weighted correction: DR, direct gradient and harmonic:0.1 weights (working default); shrink:100 is the standard comparison; legacy and global SNDR and the log trick are kept for reproducibility	DR lower bound, clip:10	the method: reward model + propensities

ARM	TRAINING OBJECTIVE	SELECTION SCORE	WHAT IT ISOLATES
No-propensity DEFAULT	naive: $-\text{mean } r \cdot \pi(a x)$	naive mean $r \cdot \pi$	learning from clicks without correcting exposure
DM-only OPT-IN	$-\sum \pi(a x) \cdot \hat{q}(x,a)$	$\hat{q}$ (all weights 0)	what a good reward model does alone
Tempered logger OPT-IN	none: logger's logits $\times s$, s in 0.5–64	DR lower bound	sharpening without any correction

PRIOR-WORK BASELINE

CausE: randomized traffic instead of propensities

CausE (Bonner & Vasile, RecSys 2018) learns a click model from the logged traffic plus a small uniform-random sample, with a control and a treatment copy of every item tied by an L1 discrepancy. The comparison asks one question. Given the same source representation and the same total budget N of target interactions, is it better to spend pN of them on uniform traffic (CausE) or to keep all N from the logger and use its propensities (OPC)? Development stage, 25k, branch `cause-fair` (`docs/cause_fair_comparison_25k.md`).



VARIANT	STARTS FROM	CAPACITY	STRUCTURAL CEILING	ROLE
Native CausE (prod-C, prod-T, avg)	Xavier-random vectors; never sees the source vectors	free user vectors, two item tables, per-row biases	the target-best value (its class contains the true click model)	the published method as it is (M5 rows, reused)
CausE-warm (C, T)	OPC's source vectors: $U \leftarrow x$, both item tables $\leftarrow a$; intercept 0 or the base rate	the native capacity, everything trainable	the target-best value	the source representation's effect at CausE's capacity
CausE-capacity-matched (C, T)	the frozen source vectors with OPC's maps at zero	$(I + D)x + b$ per side: users, control items, treatment items	OPC's Stage 1 linear-repair oracle	random vs propensity traffic at equal capacity; its greedy family equals OPC's (a test checks equal logits)

Every variant is trained by CausE's released objective (mean cross-entropy, L2, the tie; momentum with linear decay) and selected by validation NLL. The greedy (ranking) value is primary. For the stochastic value a click model's softmax has no natural temperature, so each selected model gets the tempered logger's mechanism: logits $\times s$, with s chosen by the DR lower bound (clip:10) on validation and $\hat{q}$ fit on CausE's own N rows. Search spaces were fixed on tuning seeds 200/201 before the main grid; 20 trials per variant, p and world, as OPC has per world.

```
models/cause.py (CausEModel, warm_start_, CausELinModel, batched training) ·
training/cause_trials.py · cause_trainer_trial(--methods cause --cause-family
native|warm|cap) · training/analyze_cause_fair.py (tuning and comparison tables, figures).
```

PRIOR-WORK BASELINE

BLOB: a Bayesian bandit layer on the source representation

BLOB (Sakhi, Bonner, Rohde & Vasile, KDD 2020) learns user and item representations from organic sessions, then fits a Bayesian click model on bandit feedback whose prior ties it to those representations. It never uses propensities. The controlled question: given *the same* source representation and the same N target interactions, how does BLOB compare with the likelihood learner in OPC's class (CausE-cap at $\rho = 0$), DM-only and OPC? Development stage, 25k, branch `blob-controlled-integration` (`docs/blob_controlled_integration.md`); the prior calibration on branch `blob-prior-calibration` (`docs/blob_prior_calibration.md`).

The supplied source

$\omega_u = x_u / \text{RMS}(x)$
 $\Psi = a$ (item vectors)
the logger's own vectors

BLOB's bandit layer (released graph, ported and checked)

$\beta_a = s+(w_a) \cdot \Psi_a + s+(w_b) \cdot (\Psi \zeta L^T)_a$ logit = $\omega^T \beta_a + \kappa_a + w_c$
priors: $w_a, w_b \sim N(\mu, 1); \zeta \sim N(0, I); \kappa_a \sim N(w_c, \sigma_{\kappa}^2)$; mean-field VI, Adam
N warm rows, no propensities · selected by validation NLL · greedy = $\arg\max \hat{\omega}^T \beta_a + \kappa_a$

OPC on the same vectors and rows

$((I + D)x + b) \cdot ((I + D')a + b')$ trained on DR with $\pi\theta(a|u)$
selected by the DR lower bound (clip:10) on the same 20,000 validation rows

BLOB-supplied-source replaces BLOB's organic model by the logger's own vectors, so BLOB gets no source information beyond OPC's. Both rank items with a $K \times K$ bilinear map of the same vectors; BLOB adds per-item intercepts (with a tight prior) and has no per-user terms.

ARM (25K, THE SAME SOURCE AND ROWS)	BIASED WORLDS (24)	NO BIAS (6)	WARP	GROUP	VECTOR	COMBINED
BLOB-NQ (supplied source, no propensities)	+1.42 [+0.85, +1.98]	-0.17	+1.58	+0.70	+0.84	+2.56
BLOB-MNQ	+0.94 [+0.53, +1.35]	-0.11	+0.71	+0.54	+0.75	+1.75
BLOB-Pnorm-NQ, $P_0 = 10$ (catalog-normalized prior; not the published method)	+2.82 [+1.84, +3.79]	-0.21	+2.99	+1.18	+1.06	+6.03
CausE-cap-C, $\rho = 0$ (penalized likelihood, OPC's class)	+3.09 [+2.10, +4.08]	-0.55	+4.23	+1.22	+1.03	+5.89
OPC (harmonic:0.1)	+2.69 [+1.84, +3.53]	-0.61	+3.03	+1.32	+0.96	+5.42
DM-only (own range)	+2.15 [+1.30, +3.01]	-0.55	+2.09	+0.81	+0.64	+5.07

Greedy value minus the logger's greedy value, CTR points (mean over worlds; 95% CI for the pooled biased worlds). BLOB-NQ – CausE-cap: -1.67 [-2.29, -1.06], higher in 2 of 24 biased worlds; BLOB-NQ – OPC: -1.27 [-1.77, -0.76], 2 of 24; without bias BLOB-NQ – CausE-cap: +0.38 [+0.20, +0.56], 6 of 6. BLOB-Pnorm-NQ – BLOB-NQ: +1.40 [+0.81, +1.98], 24 of 24; – CausE-cap: -0.28 [-0.57, +0.01], 8 of 24 (warp -1.24, 0 of 6); – OPC: +0.13 [-0.08, +0.34], 17 of 24; – DM-only: +0.66 [+0.30, +1.03].

BLOB's released code reproduces its paper's Table 3 (every BLO and BLOB entry within 0.02 CTR points). Here BLOB-supplied-source gets the logger's own vectors as its organic representation and the same 25k warm rows; its search space came from a pre-registered rule on seeds 200/201, with one supplementary round; selection is by validation NLL. Ceilings come from truth-trained class oracles: the value oracle (the class's best policy) and the likelihood oracle (the infinite-data likelihood fit under the logger).

The catalog-size prior

BLOB's target correction is the $K \times K$ matrix $\Delta M = s+(w_b) \zeta L^T$, added to the source map $s+(w_a) I$. The released code normalizes Ψ 's columns and then computes $L = \text{chol}(\Psi^T \Psi / P)$, so the $1/P$ normalizes a second time. Every entry of ΔM then has prior variance $s+(w_b)^2 s_{\zeta}^2 / P$, exactly: at our $P = 3,533$ –10,803 items the prior is 35–108 times tighter than at the paper's $P = 100$. **BLOB-Pnorm** computes L at a fixed reference size, $L_{P_0} = \text{chol}(\Psi^T \Psi / P_0) = \sqrt{P/P_0} \cdot L$, which makes the prior independent of the catalog ($P_0 = P$ is the published BLOB). The derivation was committed before any run.

PRIOR (NQ, TUNING WORLDS)	SCORE	– RELEASED	CORRECTION $\ \Delta M\ $ / SOURCE	USERS ON THE LOGGER'S TOP ITEM
released (L at P)	+1.79	—	0.03	60%
$P_0 = 1,000$	+2.85	+1.06 [+0.42, +1.70]	0.11	52%
$P_0 = 100$ (derived)	+3.52	+1.74 [+0.78, +2.70]	0.24	43%
$P_0 = 10$ (chosen)	+3.65	+1.86 [+0.84, +2.88]	0.28	42%

PRIOR (NQ, TUNING WORLDS)	SCORE	– RELEASED	CORRECTION $\ \Delta\mathbf{M}\ $ / SOURCE	USERS ON THE LOGGER'S TOP ITEM
prior only: s_{ζ} at $P_0 = 100$, released L (diagnostic)	+3.48	+1.70 [+0.73, +2.66]	0.21	44%

The pre-registered calibration on the 18 tuning worlds (seeds 200/201; 30 configurations per world). Score: a simulated 20-trial study selected by validation NLL, greedy gain over the logger's greedy value, CTR points. Every configuration was trained under every variant with the same batch order and noise draws. The rule kept $P_0 = 100$ unless another anchor beat it by 0.10 or more; $P_0 = 10$ did, by 0.12 [0.05, 0.20]. The prior-only variant scores like $P_0 = 100$ (−0.04 [−0.12, +0.04]), so the effect is the prior, not the optimizer's step size. Extending lr one step (to $3e-1$) moved the released BLOB by +0.05.

On the 30 main worlds, run once, BLOB-Pnorm-NQ ($P_0 = 10$) adapts as the derivation predicts. Its correction is 22% of the source term instead of 2.5%. It moves 52% of users off the logger's top item instead of 36%, and the true CTR at those moved picks is the highest of the learners (22.4%). Its click model fits better (AUC 0.697 against 0.688), and its training gap to the class ceiling falls from 4.72 to 3.56 points. Without bias it stays as close to the logger as default BLOB (81% of users kept; −0.03 [−0.34, +0.28]). Outcome B, with C's condition partly met: it is competitive, level with OPC, and tied strongest under combined bias.

`models/blob.py` (BlobBandit, BlobBanditBatch, TF1 Adam; checked step by step against the unmodified TensorFlow graph; a per-trial L scale for the catalog-normalized prior) · `training/blob_trials.py` · `blob_trainer_trial` (—methods blob —blob-families nq mnq; —blob-variants released L10 ...) · `training/class_oracles.py` (truth-trained value and likelihood oracles per score class) · `training/policy_diagnostics.py` (pick diagnostics) · `training/analyze_blob.py` (tuning rule, prior calibration, comparison tables and figures) · `scripts/blob_reference/` (the TF fixture and the Table 3 reproduction).

AUDIT, 6 OCT

What each compared arm optimizes

Read from the executable code at the commit of each reported run (`docs/training_objectives_audit.md`). All three arms start from the same frozen biased vectors, train on the same 25,000 logged rows of the biased logger (no randomized rows) and select among 20 trials on the same 20,000 validation rows.

PROPERTY	BLOB-NQ	CAUSE-CAP, $p = 0$	OPC
learned	one global $K \times K$ map $\mathbf{M} = \mathbf{s} + (\mathbf{w}_a)\mathbf{I} + \mathbf{s} + (\mathbf{w}_b)\mathbf{Z}\mathbf{L}^T$ on the column-normalized source, an intercept κ per item and a global one (variational means and stds)	$(\mathbf{I} + \mathbf{D})\mathbf{x} + \mathbf{b}$ maps for users, control items and treatment items; scale α ; intercept	$(\mathbf{I} + \mathbf{D})\mathbf{x} + \mathbf{b}$ maps for users and items; logit scale s
ranking degrees of freedom	$K^2 + P$ (κ heavily shrunk)	$K^2 + K = 1,056$ (the user map alone spans the class)	$K^2 + K = 1,056$
data-fit objective	expected logistic NLL under the variational posterior	logistic NLL of the logged clicks	harmonic-weighted DR value of its softmax policy: $\hat{q}$ term + $g(\pi/\pi_0)(r - \hat{q})$, direct gradient
propensities	no (only the post-hoc tempering of the stochastic metric)	no (same)	yes: training weights $g(\pi/\pi_0)$, selection $\min(\pi/\pi_0, 10)$
source anchoring	a Gaussian prior centred on the scaled source map; precision on the correction $\propto P$	weak: L2 on every map toward the source (28/30 selected models); at $p = 0$ the one-way L1 tie pins the item map to the source (18/30); neither restricts the class	none explicit: starts exactly at the logger; gradient clipping at 1, a fixed 5–30 epochs, weights bounded at 10 and selection by the DR lower bound act implicitly
selection	validation NLL	validation NLL	95% DR lower bound, clip:10

So the CausE-cap vs OPC contrast is a penalized click likelihood against a DR value objective in the same policy class and data, but it also changes the selection rule, the optimizer and the search spaces. At $p = 0$ CausE's own mechanism (a treatment representation learned from randomized rows) is absent: CausE-cap there is a capacity-matched penalized-likelihood baseline in CausE's objective form. OPC's $\hat{q}$ uses diagonal interactions ($\mathbf{x} \odot \mathbf{a}$) and cannot represent a general warp, which its policy class can; whether that is why CausE-cap's lead over OPC sits under warp bias (−1.19 [−1.44, −0.95], 0 of 6 worlds) is untested.

OBJECTIVE SPECTRUM, 7 OCT

Three training objectives on one correction model

The CausE-cap vs OPC contrast changed the objective, the selection rule, the optimizer and the search space at once. This study holds everything but the objective fixed: OPC's own policy model,

the same frozen source vectors, the same 25,000 logged rows and 20,000 validation rows, the same regularizer and λ grid, the same optimizer, and one paired random search (trial k is the same configuration, seed and batch order in every arm). Development stage, branch `representation-mismatch-research-next` (`docs/shared_objective_study.md`, pre-registered in \$0–\$10 before any result).

ARM	TRAINING OBJECTIVE (MINIMIZED)	NATIVE SELECTION
<code>shared_likelihood</code>	$(1/n) \sum \ell(r, \sigma(s \cdot g/T + c)) + \lambda R(\theta)$	validation NLL
<code>shared_iw_likelihood</code>	$(1/n) \sum w \ell(r, \sigma(s \cdot g/T + c)) + \lambda R(\theta)$, $w = 1/(P \cdot p)$: toward the uniform action distribution, raw	validation IW-NLL, same weights
<code>shared_iw_likelihood_clip10</code>	the same with w clipped at 10 (the one pre-specified variant)	validation IW-NLL, clipped
<code>shared_opc</code>	OPC's DR value objective (harmonic:0.1 weights, cross-fitted $\hat{q}$, direct gradient) $+ \lambda R(\theta)$; at $\lambda = 0$ the historical OPC arm exactly	DR lower bound of π_θ , clip:10

The shared model

$u' = (I + D_u)x + b_u$, $a' = (I + D_a)a + b_a$, $g = \langle u', a' \rangle$; policy $\text{softmax}(s \cdot g/T)$ with $s = \exp(30\theta_s)$ and the logger's T ; the likelihood arms read $s \cdot g/T + c$ as click logits (c changes no ranking). $2K^2 + 2K + 1$ (+1) parameters, 2,113 at $K = 32$. Its click logit spans the affine-bilinear class of the class oracles.

One regularizer

$R(\theta) = E_u \|u' - x\|^2 / E_u \|x\|^2 + E_j \|a' - a\|^2 / E_j \|a\|^2$, uniform over the catalog, in closed form $\text{tr}(DSD^T) + 2b^T D \mu + \|b\|^2$. Scale-free; s and c are not penalized. $\lambda \in \{0, 0.001, 0.01, 0.1, 1\}$ for every arm, searched per trial.

Three selections and four optima

Per world and arm: the native choice, the common choice (the 95% DR lower bound of the trial's greedy policy, the same $\hat{q}$ and rows for every arm) and the best of the 20 trials by true value (diagnosis only). The population optima θ_{log}^* , $\theta_{\text{uniform}}^*$, θ_{clip10}^* and θ_{value}^* on the affine-bilinear class (exact expectations) split each arm's shortfall into objective mismatch, training gap and selection gap.

```
models/shared_objectives.py · training/trainer_trials.py ·
regression_trainer_trial(shared_objective=...) · training/class_oracles.py
(uniform_likelihood, clip10_likelihood) · training/analyze_shared_objectives.py
(tune, oracles, compare)
```

Results (30 development worlds, 25k; greedy CTR points over the logger, 24 biased worlds)

ARM	NATIVE	COMMON DR	BEST OF 20	OBJECTIVE MISMATCH	TRAINING GAP	SELECTION GAP
<code>shared_likelihood</code>	+2.98	+2.89	+3.09	1.42	2.11	0.12
<code>shared_iw_likelihood</code>	−0.67	+0.26	+0.41	2.04	4.17	1.08
<code>shared_iw_likelihood_clip10</code>	+0.38	+0.60	+0.69	1.83	4.10	0.32
<code>shared_opc</code>	+2.66	+2.67	+2.81	0	3.81	0.15

The last three columns split each arm's distance from the class's value optimum θ_{value}^* : the population optimum of its own objective, its best trial, its native choice. The global class itself stops 2.80 points short of the target-best ceiling (1.93 group, 3.51 vector, 5.64 combined; 0.12 under warp). The four arms trained on identical rows in all 30 worlds; the shared OPC arm equals the historical one (−0.03), the shared likelihood CausE-cap (−0.12).

Step by step through one condition

Order of execution inside `training/run_full_study.py` · `_run_condition` and `training/trainer_trials.py` · `regression_trainer_trial`. The parallel and H1 runners call the same function.

#	WHAT HAPPENS	WHERE
1	Seed and pin. Deterministic kernels, fixed CPU threads, <code>seed_everything(seed)</code> ; every later stream uses <code>derive_seed(seed, labels...)</code> .	<code>utils/seeding.py</code>
2	Build the world. Calibrate once per dataset and seed (bias strengths ϵ , spread temperature T , click model α and b), then for this configuration mix the bias into the logger's view and sharpen the logger.	<code>calibrate_world</code> · <code>build_world</code> · <code>sharpen_logger</code>
3	Simulate and split. One log per train size (reg + train + 20,000 validation rows), cut into disjoint slices, cached and shared by every arm.	<code>LazyRegressionSplitCache</code>

#	WHAT HAPPENS	WHERE
4	Reward model. Fit on each size's training rows. In the default mode it is cross-fitted by user in 5 folds: each fold's model sees the other folds' rows. The external 50k-row slice is still drawn, so splits match the older mode.	<code>fit_shared_regression_bundle</code> · <code>_crossfit_bundles</code>
5	Initial reward. Exact true value of the logger, and of its greedy ranking.	<code>_dataset_log_constants</code> · <code>calc_greedy_reward</code>
6	Optuna trials per arm. Each trial reseeds, samples lr, epochs, batch size and decay, and builds a <code>CFModel</code> on the frozen biased vectors with the chosen correction. It then trains with the arm's loss, where the training loss reads the cross-fitted $\hat{q}$.	<code>regression_trainer_trial</code> · <code>CFModel</code> · <code>train</code>
7	Optional post-hoc tempering of the trained policy (logits $\times$ s chosen on validation).	<code>_post_temper</code>
8	Score on validation (logged data only) with the arm's selection score; Optuna maximizes the lower bound. The trial also logs its true value, its greedy value, the weight diagnostics and the 14 re-selection scores. None of these is used to select.	<code>_split_dr_vec_and_ess</code> · <code>_selection_score_variants</code>
9	Best trial kept (no retrain). With TPE (the default), its hyperparameters are enqueued as the first trial at the next train size. With <code>--sampler random</code> there is no warm start, so every run of the grid trains the same configurations with the same seeds (paired comparisons).	<code>study.enqueue_trial</code>
10	Write outputs for every arm.	<code>_finalize_summary_df</code>

Key data objects

NAME	WHAT IT IS	USED FOR
<code>emb_x</code> , <code>emb_a</code> TRUTH	clean vectors (with a popularity column when popularity is on)	the click model; the oracle reward model
<code>our_x</code> , <code>our_a</code>	the biased view	the logger, the policy's frozen starting vectors, the reward model's features
<code>policy_temperature</code>	the logger's sharpened temperature T/f	the logger and every learned policy (which start at the logger)
<code>pscore</code>	$\pi_0(a u)$ of the logged action, exact	importance weights
<code>q_x_a</code> TRUTH	dense true $q(u,a)$ when $\leq 25M$ cells	exact values, reward draws
<code>RegressionScoresLookup</code> , <code>CrossFitScoresLookup</code>	$\hat{q}$ rows for batches: dense or closed form; cross-fitted per user	DM terms in training and selection
<code>world</code> record	every calibrated number: ε per type and level, signal kept, α , β , T, the logger's sharpness, effective items, greedy CTR	<code>run_meta.json</code> , summary columns

Entry points and the main flags

COMMAND	PURPOSE
<code>python -m BPR.generate_artifacts</code> · <code>BPR.evaluate</code>	fit BPR v2 per dataset (factors + item bias); test-set Recall, NDCG and MPR
<code>python -m training.run_full_study(_parallel)</code>	the study; the parallel runner pools conditions across workers and GPUs
<code>python -m training.characterize_world</code>	print a dataset's calibrated world per bias configuration
<code>python -m training.oracle_repair</code>	the oracle repair bound: the learner's policy class fit to the true click model (structural recoverability)
<code>python -m training.oracle_validation</code>	checks the oracle bounds: the same class, past the old learning-rate edge, with a longer budget
<code>python -m training.analyze_recoverability</code>	recoverability tables: <code>stage1</code> (oracle) and <code>stage2</code> (learned arms as a fraction of the oracle repair)
<code>python -m training.representation_report</code>	every figure and table of the experimental report, from the committed summaries
<code>python -m training.class_oracles</code>	truth-trained value and infinite-data likelihood oracles for the affine-bilinear, BLOB and bilinear score classes
<code>python -m training.policy_diagnostics</code>	pick diagnostics of saved policies: where they recommend relative to the logger, the value there, a click model's error by logging propensity
<code>python -m training.analyze_shared_objectives</code>	<code>tune</code> (the shared arms' tuning grid and edge rule), <code>oracles</code> (the four population optima: values, losses, correction size, distances) and <code>compare</code> (the 25k comparison, the three selections and the decomposition)
<code>python -m training.analyze_blob</code>	<code>tune</code> (the BLOB tuning rule), <code>calib</code> (the prior calibration's rule and mechanism tables) and <code>compare</code> (the BLOB comparison's tables and figures)

COMMAND	PURPOSE	
<code>python -m training.run_h1_study</code>	the H1 grid (bounded-error reward model); still on the external reward model	

FLAG	DEFAULT	CONTROLS
<code>--bias-configs</code>	low medium high	levels per type, e.g. <code>high/none/none</code>
<code>--logger-greedy-share</code>	0.8	logger sharpness (<code>off</code> = spread logger)
<code>--pop-strength</code> , <code>--logger-pop-strength</code>	0, = truth	popularity in the truth and in the logger
<code>--methods</code>	opc no_propensity	add <code>dm</code> , <code>tempered_logger</code>
<code>--policy-transform</code>	linear	linear, linear+mlp, mlp
<code>--learn-logit-scale</code> , <code>--post-temper</code>	off	sharpness learned in training, or chosen after training
<code>--reward-features</code>	interaction	[x, a, $x \odot a$] or concat [x, a]
<code>--reward-data</code> , <code>--crossfit-folds</code>	train, 5	$\hat{q}$ on the policy's own rows, cross-fitted; <code>external</code> = separate 50k slice
<code>--policy-losses</code> , <code>--sn-scope</code>	dr, batch	OPC objective: dr (the default, revalidated 4 Oct); sndr with the exact scope (the gradient of the full-data SNDR ratio, direct gradient only), or the legacy batch / global surrogates (reproduction only)
<code>--lr-range</code> , <code>--epochs-range</code> , <code>--lr-decay-range</code>	1e-4 1e-3, 5 25, 0.8 1	the trained arms' shared search space; the revalidated study uses <code>--lr-range 1e-4 2e-3 --epochs-range 5 30</code>
<code>--weight-decay-range</code>	off	AdamW decay toward the logger, drawn per trial from its own stream (random sampler only)
<code>--opc-gradient</code>	direct	how OPC's loss is differentiated; <code>log-trick</code> reproduces the runs before f5cade9
<code>--train-weights</code> , <code>--select-weights</code>	harmonic:0.1, clip:10	weight transforms in training and in selection: none, clip:M, shrink: λ , harmonic: λ
<code>--sampler</code> , <code>--stage</code>	tpe, development	random = paired replay of identical configurations; the run's purpose, recorded with the results
<code>--val-size</code>	20000	fixed validation rows ($\emptyset$ = fraction rule)
<code>--methods blob</code> , <code>--blob-families</code>	off, nq	BLOB-supplied-source (opt-in; needs <code>--sampler random</code>): one row per family, <code>blob_nq</code> , <code>blob_mnq</code>
<code>--blob-lr-range</code> , <code>--blob-epochs</code> , <code>--blob-wa-m</code> , <code>--blob-wb-m</code> , <code>--blob-kappa-s</code>	released values	BLOB's search space: lr, epochs, the prior means of w_a and w_b , the intercepts' prior std (searched when several are given)
<code>--blob-seed-tag</code> , <code>--blob-pick-diagnostics</code>	off	an independent set of trials (a supplementary tuning run); every trial's pick diagnostics
<code>--blob-variants</code>	released	BLOB prior variants, each trained on the same configurations, batch order and noise draws: $L \langle P_\emptyset \rangle$ computes L at catalog size $P_\emptyset$ (BLOB-Pnorm; rows <code>blob_1<P>_nq</code>), $S \langle P_\emptyset \rangle$ puts the same prior on ζ instead (diagnostic)
<code>--methods shared_likelihood</code> <code>shared_iw_likelihood</code> <code>shared_iw_likelihood_clip10</code> <code>shared_opc</code>	off	the shared-objective arms (opt-in; <code>--learn-logit-scale</code> required, not post-tempered): one correction model, one paired search, four objectives
<code>--shared-lambdas</code> , <code>--shared-arm-space</code> , <code>--shared-arm-lambdas</code> , <code>--shared-seed-tag</code>	0 ... 1, none, none, shared	the source anchor's λ grid; one arm's own range or λ grid (the edge rule's extension; its draws map the same uniform numbers onto its range); an independent set of paired trials (a supplementary tuning round)
<code>class_oracles</code> <code>--objectives</code>	value likelihood	add <code>uniform_likelihood</code> and <code>clip10_likelihood</code> : the population objectives of the weighted-likelihood arms
<code>--save-policies</code>	off	save each arm's selected policy in its condition folder, for <code>training.policy_diagnostics</code>
<code>--skip-completed</code>	on	run only what a condition folder lacks: a requested arm is done when each of its labels has a row for every requested train size made with the same settings (<code>arm_config_key</code>); a partly done CausE or BLOB arm runs only its missing rhos or variants; rows made with other settings stop the run before anything starts. <code>--no-skip-completed</code> reruns the requested arms and replaces their rows; other arms' rows are always kept

The runs before 26 Sep reproduce exactly with `--reward-features concat --train-weights none --`

`select-weights clip:1 --policy-transform mlp --logger-greedy-share off --reward-data`

`external --val-size 0` (checked end to end against ce827f5).

Outputs and what the columns mean

COLUMN	MEANING
<code>policy_rewards</code> TRUTH	true CTR of the selected policy; at train size 0, the logger's
<code>policy_rewards_greedy</code> , <code>actual_reward_greedy</code> TRUTH	true CTR of the policy's top item per user: rankings compared without sharpness
<code>actual_reward</code> TRUTH	each trial's true CTR (never used to select)
<code>value</code> , <code>r_hat</code>	each trial's selection score (lower bound) and point estimate
<code>sel_ci_low[spec]</code> , <code>sel_r_hat[spec]</code>	re-selection scores under 14 rules, every arm
<code>ess</code> , <code>ess_raw</code> , <code>diag_*</code>	effective sample size of the transformed and raw weights; weight profile; DM vs correction split
<code>logit_scale</code> , <code>post_scale</code>	learned and post-hoc sharpness factors
<code>trial_time_s</code> , <code>diverged</code> , <code>param_weight_decay</code>	each trial's wall time; whether training stopped at a non-finite gradient (the trial keeps its last finite parameters); its AdamW decay
<code>code_commit</code> , <code>search_space</code> (run_meta.json, manifest)	the commit and dirty flag of the code that ran; the search space
<code>logger_greedy_share</code> , <code>logger_sharpness</code> , <code>logger_greedy_ctr</code>	the logger of the condition
<code>reward_data</code> , <code>crossfit_folds</code> , <code>post_temper</code> , ...	the settings of the run, one column each
<code>diag_dr_greedy_low</code> , <code>diag_anchor_R</code> , <code>diag_changed_share</code> , <code>diag_changed_gain / _loss</code> , <code>diag_val_nll</code> , <code>diag_val_iw_nll</code> , <code>param_anchor_lambda</code> (shared arms)	per trial: the common selector (the 95% DR lower bound of its greedy policy), the correction size $R(\theta)$, the share of users whose top item differs from the logger's and the value gained and lost there, the validation NLL plain and weighted, the anchor's λ
<code>train_rows_sha1</code> , <code>train_click_sum</code> , <code>train_w_ess</code> , <code>train_w_q99.9</code> , <code>head_logit_scale</code> , ... (shared arms)	per size: the data identity of the training and validation rows, the uniform-reference weights' profile (ESS, quantiles, share clipped at 10) and the likelihood head's starting point
<code>arm_config_key</code> (summary); <code>labels</code> , <code>arm_configs</code> (run_meta.json)	a hash of the settings that produced the row, the completion marker of <code>--skip-completed</code> ; each label's arm, configuration, code commit and time

Condition folders name every non-default setting, e.g.
`dataset=m1__bias=medium__ctr=0.05__seed=100__qhat=train__cf=5__val=20000` (`__mix=` for a logger mixed with uniform). A folder collects rows from any number of invocations; `run_invocations.jsonl` at the run root records each one (`run_manifest.json` holds the latest).

Modules

utils/

- `representation_bias.py`: calibration, the three bias types, logger sharpness, world record
- `simulation_utils.py`: click model, `generate_dataset`, logging, exact and greedy values
- `importance_weights.py`: none, clip, shrink, harmonic, dm; ESS
- `policies.py`, `seeding.py`, `noise_snr.py`, `provenance.py` (the code commit of a run)
- `budget_split.py`: CausE's fixed budget, $(1 - p)N$ logger rows + pN uniform rows

models/

- `models.py`: CFModel (logit scale, popularity weight), corrections, `RegressionModel`
- `custom_losses.py`: snr (batch or global scope), dr, dm, ipw, naive, kl, crm; log-trick or direct
- `estimators.py`: post-hoc DM, DR, SNIPW, SNDR (clip, shrink, harmonic)
- `cause.py`: CausE (prod and avg layouts, the released objective, checked against the TensorFlow graph), `warm_start`, `CausELinModel`, batched trials
- `shared_objectives.py`: `SharedCorrectionModel` (OPC's CFModel with a click intercept in click mode), `SourceAnchor` $R(\theta)$, `LikelihoodLoss` (plain or weighted $1/(Pp)$, raw or clipped), `AnchoredPolicyLoss`, the click head's fit and the weight profile
- `blob.py`: BLOB's bandit layer (NQ and MNQ, the released priors, Ψ normalization, noise term and TF1 Adam; checked step by step against the unmodified TensorFlow graph), batched trials, a per-trial scale on L (the catalog-normalized prior)

training/

- `trainer_trials.py`: splits, reward models, cross-fitting, trials, selection, tempering
- `run_full_study(_parallel).py`: conditions, arms, flags, outputs; both runners plan, run and merge a condition with the same functions
- `run_state.py`: which rows a request expects, whether a condition folder holds them (`--skip-completed`), and merging a run's rows into the folder (no duplicates, no dropped arms)
- `oracle_repair.py`: the oracle repair bound; `oracle_validation.py`: its check; `analyze_recoverability.py`: recoverability, per-dataset and gap tables
- `memory_budget.py`: workers per GPU from each condition's estimated peak and the free memory
- `representation_report.py`: the experimental report's figures and source tables
- `cause_trials.py`: the CausE arm (native, CausE-warm, CausE-capacity-matched; fixed-budget data, seeded search, NLL selection, the fair tempering); `analyze_cause_comparison.py`, `analyze_cause_fair.py`: the M5 and the fair comparison's tables and figures
- `blob_trials.py`: the BLOB-supplied-source arm (the logger's vectors as BLOB's source, seeded batched search, NLL selection, the fair tempering, paired prior variants); `class_oracles.py`: truth-trained oracles per score class and objective; `policy_diagnostics.py`: pick diagnostics of saved policies; `analyze_blob.py`: the BLOB tuning rule, the prior calibration and the comparison; `analyze_shared_objectives.py`: the shared arms' tuning rule, the population optima and the comparison
- `logging_coupling_diagnostic.py`: the exact distribution the pre-fix logger generated; `analyze_revalidation.py`, `revalidation_phase3.py`, `revalidation_compare.py`, `revalidation_report.py`: the revalidation's tuning analysis, summaries, old-vs-new pairing and report
- `training_utils.py`, `characterize_world.py`, `run_h1_study.py`

BPR/

- `bpr.py`, `generate_artifacts.py`: BPR v2 with item bias, numpy trainer
- `evaluate.py`: test-set metrics on CPU and GPU
- `bpr_config.py`: per-dataset recipes and the default six datasets

Tests

646 tests pass with the GPU; 616 pass and 11 are skipped with it hidden (at 24a4afa, branch `representation-mismatch-research-next`). Every change since 25 Sep was also checked bit for bit against the previous commit with the new option off. New on 7 Oct, with the shared-objective study:

- the shared model and OPC's `CFModel` give identical logits, policies, rankings, exported vectors and OPC-loss gradients for identical parameters; in click mode the logits are $s \cdot g / T + c$;
- `shared_opc` searching only $\lambda = 0$ reproduces the historical OPC arm trial for trial (every logged column exactly);
- $R(\theta)$ against the direct mean displacement, and its invariance to rescaling the source; the plain, weighted and clipped likelihoods and the anchored policy loss against manual calculations; the weights $1/(P \cdot p)$ and their profile; the click head's fit;
- the four arms draw the same configuration for trial k , train on the same rows (hash, clicks, propensities) and select natively; an arm with its own range keeps the pairing by quantile; a new seed tag draws new paired trials; the runner refuses to post-temper the shared arms;
- the oracles' weighted objectives; the analysis on synthetic trials: the three selections, the edge rule, pooling a supplementary round, the population tables and the decomposition adding up.

New on 6 Oct, with the review's fixes:

- one condition collects, over several invocations of the real command lines, OPC, DM-only, two CausE families and two BLOB prior variants, with an invocation that dies after training and its resume: each label once per train size, each trial once, completed rows unchanged bit for bit, a repeated request changes no file, and the folder equals the same arms run at once; the parallel runner and its worker pool do the same;
- the configuration key, the status of a folder, the narrowing to missing rhos and variants, the merges (kept rows keep their digits) and the per-arm log reset;
- the world characterization builds the study's worlds; the pick diagnostics rebuild a mixed logger; a dense policy has the value of the same policy object; the slim run log describes the rows the run trained on; the OOM detector, the manifests, the parse-time loss check, the help defaults, and the CausE diagnostics without autograd. These tests fail on the code before the fixes.

Every committed table of the OPC revalidation, the fair CausE comparison, the controlled BLOB comparison and the prior calibration was regenerated from the run folders with the fixed code: every value is identical (aside from the corrected CausE-cap label, and a `variant` column that the 5 Oct code added to the BLOB tuning tables), and so are the pick diagnostics of all 30 worlds.

New on 5 Oct, with the BLOB prior calibration:

- per-trial L scales in the batched layer equal separate single runs (NQ and MNQ); scaling L by c is the same correction as scaling ζ by c ;
- paired prior variants share their noise stream, and adding variants leaves the released arm's trials unchanged;
- the calibration rule's choice and its tie-break toward the derived $P_0 = 100$.

New on 5 Oct, with BLOB:

- the BLOB port replays the authors' unmodified TensorFlow graph step by step in five cases (losses to $2e-5$, variables to $1e-4$, the released initialization exactly); the released Ψ normalization aliases the prior mean; the fixture's provenance is the unmodified release;
- batched BLOB trials equal separate runs, and the sync-free training loop is bit-identical to the first one (a diverging trial included);
- the BLOB arm end to end, deterministic, with the supplied source at unit RMS;
- every class oracle starts at the logger; the ranking vectors and the click offset reproduce the scores;
- the pick diagnostics against dense numpy; every arm's saved policy (OPC, CausE-cap, BLOB) reproduces its own greedy value;
- the BLOB tuning rule on synthetic trials (the simulated protocol, the decision, the edge rule).

New on 5 Oct, with the CausE comparison:

- CausE's released objective against the unmodified TensorFlow graph; batched trials equal their single runs;
- CausE-warm starts exactly at the source vectors; CausE-capacity-matched gives the same logits as OPC's policy for the same maps ($\alpha = s/T$); at $p = 0$ only the symmetric tie moves the treatment side;
- tempering only the selected trial reproduces tempering every trial; `--skip-completed` recognizes every CausE family's labels (exactly, since 6 Oct);
- the fixed budget: $N_c + N_t = N$, the warm rows are a prefix of OPC's split, the uniform rows a separate draw;
- the fair comparison's analysis: the resampled k-trial protocol equals direct selection when k covers every trial; paired contrasts by world.

New on 4 Oct, with the simulator fix:

- logged rows are draws from the propensities they record: per fixed user (chi-square at three temperatures, with and without a uniform mix), jointly over users and actions, and through the IPS identities $E[\pi/pscore] = 1$ and $E[\pi/pscore \cdot r] = V(\pi)$. 13 of these 15 tests fail on the pre-fix code;
- the smallest failing case: two users, two actions, $\pi_0 = (\frac{1}{2}, \frac{1}{2})$; with the coupled streams user 0 always got action 0;
- a sharper logger is more concentrated in the population and in the logs (repeat rate $= \Sigma \pi_0^2$);
- the search space: the default ranges reproduce the run without one, a new range changes only its own draws, and AdamW decay comes from its own stream so trials stay paired;
- a diverging trial keeps its last finite parameters; exact SNDR equals the literal full-data ratio's gradient;
- the revalidation report rebuilds from the committed summaries alone, and its old-vs-new numbers are paired per-world differences.

Earlier (27 Sep):

- each objective variant's gradient against literal full-data autograd;
- equal row weights across the DataLoader's short final batch;
- identical replayed configurations;
- log-trick and direct gradients for every transform;
- the harmonic transform: formula, limits, the gradient bound and serialization;
- the oracle repair: the learner's class, starting exactly at the logger, deterministic; the recoverability tables regenerate byte for byte;
- a one-size run reproduces that size's trials of a multi-size run, which is what lets slices pair with the main runs;
- the worker planner against what a study run holds: dense $\hat{q}$ copies during training and at their peak, and plans for one, two and four GPUs;
- the oracle validation reproduces the Stage 1 winners exactly and extends past a winning edge; the follow-up tables never lower a bound and decompose the target gap exactly.

The world

- calibration targets on the full population; nested levels; seeds
- logger sharpness hits its share exactly; only the logger changes
- popularity column; exact and greedy values on CPU and GPU

Learning and selection

- device scoring equals the numpy reference, CPU and GPU
- weight transforms; DM loss; logit scale; corrections start at the logger
- cross-fitting: folds never see their users, fast forms equal routing

The study

- every arm runs and logs its scores through the study path
- reward-data modes, run keys, CLI defaults, skip rules
- end-to-end reproducibility per seed

Open risks

Found by reading the code; each is a hypothesis for a targeted test. Fixed items are kept, faded, for the record. The first four come from the 6 Oct review of the live workflows (docs/handoff_20261006.md §5). None changed a reported number, and all were fixed on 6 Oct, except the memory planner's high estimate for CausE- or BLOB-only runs, kept as a conservative bound.

■ may change main results ■ post-hoc or robustness ■ reporting

The world-characterization command crashed (26 Sep–6 Oct). FIXED training/characterize_world.py · characterize

The world options gained `logger_greedy_share`, which `WorldConfig` did not accept, and the share never reached the world. Each world is now built through `generate_dataset`, exactly as the study builds it.

Re-running a condition in its own folder repeated its trials, and a partial rerun dropped other arms' rows. (FIXED) `run_state · run_full_study · execute_condition`

An arm now drops its own rows for the sizes it is about to run. Every file of the folder is merged: a run replaces exactly the (label, train size) rows it produces. Every loader keeps one row per trial. One run of 4 Oct holds identical duplicated trials (two conditions of `run_reval_sndr_exact_harm0.1_s201`); it was left as it is.

--skip-completed counted any CausE or BLOB row as the whole arm. (FIXED) `run_state · condition_plan`

Completion is now checked per label, train size and configuration key. Another CausE family or BLOB variant runs next to the existing ones, and rows made with other settings stop the run before it starts.

Smaller review findings. (FIXED)

Each of these is fixed:

- the CausE trial diagnostics run without autograd again;
- the pick diagnostics rebuild the logger with its uniform mix, and a mixed logger gets its own folders;
- a dense policy's value weights users by the prior;
- the slim run log uses the rows the run trained on;
- the OOM detector matches whole words;
- one manifest builder serves both runners;
- the parallel `--policy-losses` is checked at parse time;
- two help and docstring defaults are corrected.

Kept on purpose: the memory planner still counts OPC's $\hat{q}$ copies for CausE- or BLOB-only runs. That estimate errs high, which costs throughput but not results; lowering it needs measured peaks.

The logger reused its users' random draws (69fffab..c11b2b3, 24 Sep–4 Oct). (FIXED) `trainer_trials · _simulate_from_embedding_policy · simulation_utils`

The logger's generator was seeded like the simulation's, and one uniform per row drew both the user and the action. Each user therefore received a nearly fixed action (93–99% of its probability), while the stored propensity was the logger's softmax probability, 6–25 times too small (20–40 at logger share 0.6). The logs had about one effective action per user at every logger share. Fixed in dbc401b: the actions come from their own stream (`derive_seed(..., "logging_policy_actions")`), and the simulation refuses a policy that shares its generator. Every affected experiment was rerun on the corrected logs (`docs/simulator_fix_opc_revalidation_20261004.md`); the regression tests above fail on the old code.

The working defaults are not the paper's choice yet. `custom_losses.py · dr_correction · docs/training_losses.md §3.4, §9`

Legacy SNDR's objective depends on the batch size. Global SNDR removes that, but it is a stop-gradient, epoch-stale surrogate. DR has no normalizer. Under the log trick, a weight transform optimizes $H(w)$ rather than the named estimate; `--opc-gradient direct` optimizes the estimate itself. All of these are implemented and characterized. Since f5cade9 (27 Sep) the working defaults are DR, the direct gradient and harmonic:0.1; on 4 Oct they were re-tuned on the corrected logs and kept, with a wider learning-rate range. The runs before f5cade9 are reproduced with `--policy-losses sndr --opc-gradient log-trick --train-weights shrink:100`. The paper's method is frozen only at the protocol freeze.

Best hyperparameters carry over between train sizes (TPE). `study.enqueue_trial`

Results at different sizes are not independent, and the order of sizes matters. `--sampler random` removes the warm start for paired comparisons.

Some defaults differ from what the studies used. `run_full_study(_parallel) · _run_condition · trainer_trials`

The CLI defaults to `--sampler tpe`, which is unpaired and warm-starts each size, so paired studies need `--sampler random`. `--learn-logit-scale` is off, the search space defaults to the older range (the revalidated study passes `--lr-range 1e-4 2e-3 --epochs-range 5 30`), the default seeds are 0/1/2 (development used 100/101), and DM-only and the tempered logger are opt-in. Direct callers of `_run_condition` get an external $\hat{q}$ with no cross-fitting unless they pass `reward_data`, and the trainer API falls back to shrink:100 weights. Pass these explicitly. The details are in `docs/roee_handoff_20260928.md`.

An oversubscribed GPU slows down silently (WSL2). `memory_budget.py` · `run_full_study_parallel`

Under WSL2, workers that ask for more GPU memory than is free do not fail. The driver spills into shared system memory: the GPU stays 100% busy at about 130 W instead of 250–290 W, every worker slows about 50×, and no error is raised. This was confirmed on 27 Sep with the Windows shared-memory counter (anime at 4 workers). The worker planner now sizes each condition from its own budget. The estimate is the training step at its largest batch + its dense $\hat{q}$ copies + 1.5 GiB. There are 3 dense copies with cross-fitting and a per-size refit, because the next size's lookup is built while the previous one is still held. Each GPU then takes $\text{floor}(0.75 \times \text{free} / \text{peak})$ workers, over any number of cards, with `--max-workers` as the cap (`f2e03b5, 05222f8`). anime now runs 2 workers on the 48 GB card with no spill. Still open: the planner sizes one run and does not see other runs on the same card. Two concurrent revalidation runs spilled on 4 Oct and were resumed with identical results; keep at most 3 anime workers in total. In GPU mode host RAM is not planned either, and an anime worker holds about 14.5 GiB. On a large multi-GPU machine RAM can bind first. Results are unaffected, because runs are deterministic.

One non-finite gradient failed the whole condition. `FIXED` `trainer_trials` · `regression_trainer_trial`

Since 4 Oct the trial stops training at the first non-finite gradient, keeps its last finite parameters and is flagged `diverged`; other errors still propagate.

Sharpening is a large part of any stochastic gain.

The tempered logger gains 2.3–6 points over the logger by sharpening alone. With the learnable logit scale OPC sharpens too. With the scale fixed and the older search range (the CausE M5 configuration) OPC is about 1 point lower at 25k on the same logs, and 2.3 lower without bias. Post-hoc tempering on top of the learned scale adds nothing to the selected policy on corrected logs (not adopted). Comparisons with other methods should report greedy values as well, or give every policy the same sharpening.

A logging uniform mix enters the initial reward but not the start point.

Only matters with `--logging-uniform-mix` > 0 (off by default).

Labels.

The post-hoc `ipw` column is self-normalized; the trial column `q_error` is the validation standard error.

Fixed this week. `FIXED`

Clicks capped at the CTR, popularity-dominated vectors and noise that inflated vector length (simulator revamp). A reward model with one ranking for every user (interaction features). Selection clipped at 1. One reward model reused across sizes and fit on extra data (budget-fair $\hat{q}$). No-propensity trials missing re-selection scores (`76a29e6`). On 27 Sep: the DataLoader's final short batch was upweighted by up to 4.8× (`b5efc7d`); `--no-log-trick` never reached OPC (documented, `3bed8de`); post-hoc estimators used raw weights for an unknown transform, which now raises (`d40aaef`).

Findings so far

All development runs (seeds 100/101), on the corrected logs since 4 Oct. The numbers and their old-vs-corrected classification are in `docs/simulator_fix_opc_revalidation_20261004.md`; the pre-fix reports are kept as historical records. The CausE comparison (5 Oct) is in `docs/cause_fair_comparison_25k.md`, the BLOB comparison (5 Oct) in `docs/blob_controlled_integration.md` and its prior calibration in `docs/blob_prior_calibration.md`; what each compared arm optimizes in `docs/training_objectives_audit.md` (6 Oct); the three objectives on one model in `docs/shared_objective_study.md` (7 Oct).

With everything but the objective held fixed, the likelihood still leads OPC at 25k, and selection is not why.

On OPC's own model, regularizer, data, optimizer and search, the penalized likelihood gains +2.98 greedy points on the 24 biased worlds and OPC's objective +2.66: +0.32 [0.10, 0.54], from warp (0.98, 6/6) and combined bias (0.32, 5/6); group and vector bias are ties. Under one common DR selector the lead is 0.22, among the best of 20 trials 0.28. OPC's trials are better on average under group, vector and combined bias (18 of 18 worlds), but not at the top. Development seeds, 25k (`docs/shared_objective_study.md` §14–§16).

OPC's population advantage is spent in training.

The likelihood's population optimum ranks 1.23, 1.12 and 3.36 points below the value optimum under group, vector and combined bias (18/18 worlds; 0 under warp), while the likelihood barely separates the two (0.0044 nats per row). At 25k OPC's best trial stays 3.81 points below the value optimum, against the likelihood's 2.11 below its own: the mismatch (1.42) and OPC's extra training gap (1.70) nearly cancel.

Weighting the likelihood toward uniform actions moves it away from value.

In the population the uniform-weighted optimum ranks 0.63 points below the logging-likelihood optimum on the biased worlds (clip 10: 0.41). At 25k the weights' effective sample size is about 39 of 25,000 rows: the raw arm loses 0.67 points to the logger (its own selection rule once lost 19), and clipping at 10 recovers only +0.38. The one tuning extension (lower learning rates) did not help.

The CausE-cap vs OPC contrast is more than likelihood vs value.

CausE-cap at $p = 0$ is a penalized likelihood in OPC's class: its L2 term shrinks every map toward the source (28 of 30 selected models), and at $p = 0$ CausE's one-way tie becomes an L1 anchor that held the item map at the source in 18 of 30. Neither restricts the ranking class, because the user map alone spans it. OPC maximizes a harmonic-weighted DR value with no explicit penalty. The two also differ in selection (validation NLL vs the DR lower bound), optimizer and search spaces, so the 0.41-point lead was not attributable to the objective alone. On 7 Oct, with all of these held fixed, the plain likelihood still led OPC's objective by 0.32 (above).

BLOB's weak adaptation was mostly its catalog-size prior.

The released prior puts variance $s+(w_b)^2/P$ on every entry of BLOB's correction, 35–108 times tighter here than at the paper's $P = 100$. With L computed at a fixed P_0 (pre-registered; $P_0 = 10$ chosen on seeds 200/201), BLOB-Pnorm-NQ gains +2.82 on the 24 biased worlds. That is +1.40 [0.81, 1.98] over the published BLOB in 24 of 24 worlds, level with OPC (+0.13 [−0.08, 0.34]), 0.28 [−0.01, 0.57] below CausE-cap (all of it under warp: −1.24, 0/6) and 0.66 above DM-only. Under combined bias it has the highest mean (+6.03; CausE-cap +5.89). Its correction grows 8.5-fold and its training gap closes (4.72 → 3.56); without bias it stays at the logger. Its remaining weakness is NLL selection (0.32 points of regret against 0.13–0.16). Development seeds, 25k.

Given the same source, the published BLOB is the weakest model-based learner at 25k.

BLOB-supplied-source (the logger's vectors as BLOB's organic representation, the same 25k rows, no propensities) gains 1.42 greedy points over the logger on the 24 biased worlds. It trails the likelihood learner (CausE-cap at $p = 0$) by 1.67 [1.06, 2.29] (higher in 2 of 24 worlds), OPC by 1.27 [0.76, 1.77] and DM-only by 0.74. Without bias it loses least (+0.38 over CausE-cap, 6/6). Its released code reproduces the paper's Table 3 within 0.02 points, so this is not a broken BLOB; it is BLOB applied where its source is wrong. The calibrated prior (above) closes most of the gap.

BLOB adapts little: training, not capacity and not the likelihood principle.

Truth-trained class oracles put BLOB's ranking class within 0.07 points of OPC's structural ceiling, and its infinite-data likelihood fit (+5.49) above the plain learner's (+5.21). About three quarters of its deficit is what its trials reach, the rest NLL selection. Its $K \times K$ deviation stays at about 2.5% of the source term, and it keeps 64% of users on the logger's top item (CausE-cap 38%, OPC 42%). OPC's lead comes from the users it moves to less-logged items (+1.11 of +1.27). The lead this pointed to, the deviation's scale $L = \text{chol}(\Psi^T \Psi / P)$ shrinking as $1/\sqrt{P}$, was tested on 5 Oct: it is the prior (above).

The likelihood objective costs value where the mismatch is not a warp.

With unlimited data the likelihood fit in OPC's class reaches its value ceiling under warp (+7.2) but gives up 1.2 (group), 1.1 (vector) and 3.4 (combined) points against it. So the likelihood learner's lead over OPC under warp is finite-sample, and the cells where a value objective can matter are group, vector and combined mismatch: the ones the planned richer corrections target.

At equal capacity, CausE's learner ranks better than OPC at 25k, and randomized traffic is not why.

CausE-capacity-matched trains OPC's own correction family on the same source vectors with CausE's objective and NLL selection. It beats OPC by 0.41 greedy points [0.16, 0.66] at $p = 0$ and 0.36 [0.15, 0.57] at $p = 0.25$ (24 biased worlds, paired), with tempered stochastic value close to that. Its lead sits in warp worlds (1.19 points, 6/6). Group and vector are ties. Uniform rows change it by at most 0.05 points, so at $p = 0$ (no randomized rows, no propensities) it is already ahead. It also beats DM-only by 0.94 points. Which part does it, the likelihood objective or the selection rule, is open. 25k only; a targeted 100k check is recommended.

CausE at its native capacity loses, mostly for lack of the source representation.

Started at OPC's source vectors (CausE-warm), CausE gains 8.3 points over native CausE, yet still trails OPC by 1.0 (C) and 2.4 (T): its free per-user vectors learn little from 25k rows (0.15 of the loss repaired; ceiling 1.0). Swapping free vectors for the linear family adds 1.4 more. Randomized traffic is a pure cost at 25k: $p = 0.25$ gives up 932 clicks per world (21% of the collection's clicks) for no gain in either fair variant.

Structure decides what a linear repair can fix.

We trained the learner's own linear class on the true click model. It recovers 0.99 of the ranking loss from warp, 0.56–0.78 from group and 0.31–0.61 from vector (high level). The order warp > group > vector held in all 6 dataset × seed pairs, and a validation with up to 9× the optimization budget moved it by at most 0.025. This uses no logged data, so the logging fix leaves it unchanged.

Logged learning recovers a fifth to a half of it.

OPC reaches 0.21 / 0.38 / 0.49 of the oracle's ranking repair at 5k / 25k / 100k (biased worlds), 5.6–7.5 points over the logger. At 100k the loss splits (structural / not learned / learned) as warp 0.01 / 0.47 / 0.52, group 0.32 / 0.39 / 0.29, vector 0.50 / 0.29 / 0.21 and combined about 0.32 / 0.30 / 0.37. No-propensity learning recovers 0.10–0.15, and sharpening alone none.

Propensities add value over a good reward model.

OPC beats DM-only by +1.22 / +0.64 / +0.77 points at 5k / 25k / 100k (biased worlds), and for every single-type bias at every size with every interval above 0. Under combined high bias the two tie below 100k; there DM-only leads on ml at 25k by 1.0 point. OPC beats no-propensity by 1.9–2.4 points from 25k.

Unneeded correction costs a little.

With no bias OPC trails sharpening alone (the tempered logger) by 0.6–1.2 points, and at 100k no-propensity by 0.4.

OPC holds up when the reward model's data shrink.

Cutting $\hat{q}$'s data from a separate 50k-row slice to the policy's own rows costs DM-only 2.15 points at 5k and OPC 0.25 (within noise). With 50k rows for $\hat{q}$, DM-only beats OPC at 5k by 1.2.

A misspecified reward model breaks DM-only, and OPC's capped weights at large n.

With concat features (one ranking for every user), DM-only loses 3.6–5.6 points: $\hat{q}$ overrates its pick by 11–17 points and its policy leaves the logged support. OPC loses 1.6 points at 25k and 4.5 at 100k with harmonic:0.1 or shrink:100: their cap at 10 lets $\hat{q}$'s errors steer training off the support, where the clipped selection estimate cannot see it. With raw training weights OPC loses nothing at 100k, at a cost of 0.8 points when $\hat{q}$ is well specified. OPC still beats DM-only.

Logger support 0.6–0.95 does not remove recoverability (25k).

The corrected loggers spread over 31–36, 7–8 and 2–3 effective actions per user at shares 0.6, 0.8 and 0.95. OPC's recovery is flat from 0.8 to 0.95, slightly lower at 0.6 for warp and group (not significant), and OPC leads DM-only at every share. A logger without full support was not tested.

The weighting defaults survive re-tuning.

On corrected logs every regularized weighting beats raw DR, the best amount of regularization falls with n, harmonic $\lambda = 0.1$ –0.3 lead, and Su's shrink:100 is within 0.2 points of harmonic:0.1 in every cell of the Stage 2 grid. Exact SNDR adds nothing over DR, and the log trick never beats the direct gradient. The learning-rate range was too narrow and is now $1e-4$ – $2e-3$, with 5–30 epochs. The weighting is regime dependent: with a misspecified reward model raw DR is the robust choice (above).

Selection works; training is the bottleneck.

The clip:10 DR lower bound picks within 0.03–0.11 points of the best of 20 trials, and OPC's selection regret is 0.09–0.14 points. Its DR estimate of the selected policy is unbiased within noise from 25k (it was optimistic on the buggy logs). DM-only's reward model overrates its own pick, by 5.3 points at 5k.

Plan

Status, 4 Oct

DONE A logging bug (69fffab..c11b2b3: each user's action reused the draw that picked the user) was found, fixed (dbc401b) and made a tested invariant. Every affected experiment was rerun on the corrected logs: Stage 2 with all arms and the shrink:100 robustness arm, the logging-support sweep, both reward-model tests, and a decomposition of the simulator fix vs the retuning. The OPC defaults were re-tuned on new development seeds (200/201): DR, the direct gradient, harmonic:0.1 and clip:10 selection are kept, and the learning-rate range is widened to $1e-4$ – $2e-3$ with 5–30 epochs. Every old finding is classified old vs corrected in `docs/simulator_fix_opc_revalidation_20261004.md`; figures and tables are in `artifacts/full_study/opc_revalidation_20261004/report/`. The earlier reports are kept as historical records, marked as produced on the buggy simulator.

CausE, 5 Oct

DONE The fair 25k comparison on branch `cause-fair` (archive `origin/cause-baseline = d31a5dd`). Native CausE (M5 rows, reused), CausE-warm and CausE-capacity-matched, at $p \in \{0, .01, .05, .10, .15, .25\}$, against the revalidated OPC. The search spaces were tuned on seeds 200/201 by a pre-registered rule. **AWAITING REVIEW** The next decision is the user's: a targeted 100k run (CausE-cap at $p \in \{0, .05, .25\}$), and whether to check OPC's objective vs selection against the in-class likelihood learner.

BLOB, 5 Oct

DONE The released code reproduces the paper's Table 3; the PyTorch port replays the released graph step by step. BLOB-supplied-source was compared at 25k on the 30 development worlds with the reused OPC, CausE-cap, CausE-warm and DM-only rows, on branch `blob-controlled-integration`. Its search space came from a pre-registered rule on seeds 200/201, with one supplementary round. **REVIEWED** The review asked for the prior calibration first (below).

BLOB prior, 5 Oct

DONE Where catalog size enters BLOB's prior was derived and committed before any run. A pre-registered calibration on seeds 200/201 (released; $P_0 = 1,000, 100, 10$; a prior-only diagnostic) chose $P_0 = 10$, and BLOB-Pnorm-NQ was run once at 25k on the 30 development worlds, on branch `blob-prior-calibration`. Outcome B, with C's condition partly met: BLOB-Pnorm is a competitive core baseline, and the published BLOB stays the reference. **AWAITING REVIEW** Propensity-weighted likelihood, richer corruption, hierarchical correction, RecoGym and 100k wait for the review.

Review, 6 Oct

DONE The training objectives of BLOB-NQ, CausE-cap and OPC were audited from the executable code. The live workflows were reviewed (14 findings, none changing a reported number), 13 were fixed and tested end to end, and every reported table was reproduced with the fixed code (`docs/handoff_20261006.md`). `handoff-20261006` is the production / handoff checkpoint; new research starts from the same commit on `representation-mismatch-research-next`. **AWAITING DECISION** The 100k CausE-cap check; separating CausE-cap's objective from its selection rule; BLOB-Pnorm as the core model-based baseline.

Objectives, 7 Oct

DONE Three training objectives on OPC's global correction model, with one source-anchoring regularizer and one paired search: penalized likelihood, uniform-weighted likelihood (raw and clipped at 10) and OPC's value objective, at 25k on the 30 development worlds, with their population optima and native, common and best-trial selection. Search spaces were pre-registered and tuned on seeds 200/201, with one extension (the weighted likelihoods' learning rate). **AWAITING REVIEW** Recommended for the correction-capacity study: penalized likelihood and OPC as the primary objectives, the clipped weighted likelihood as a reference arm, and the common DR selector reported beside every native rule. Group, regional and individual correction, new corruption, 100k and RecoGym wait for the review.

Dev grid

LATER linear vs linear+MLP vs MLP corrections for no bias, medium, high, and warp, group or vector alone (ml, kuairand, anime).

Final report	THEN all six datasets × no / low / medium / high bias × warp-only, group-only, vector-only × 5k, 25k, 100k, four arms, fresh seeds.
Model changes	PENDING bilinear reward model; per-group and per-item offsets with partial pooling; popularity runs; cluster-level weights for large catalogs; Switch-DR; validation as a share of the budget; aligning H1 with the new defaults; real-data checks (Open Bandit Dataset, KuaiRec's full matrix, KuaiRand's random exposures, Coat, Yahoo! R3).
Assumption	Logged propensities are known exactly, as in most OPE and OPL work; estimating them is outside this study. Since 4 Oct the simulator's logged propensities are tested to be the probabilities that generated each row.
Deferred	Comparing policies by their exploitation (greedy) part alone. The greedy value is logged on every trial for when we return to it.